



BEASTLY FACTS · CARE PACKAGE

BALL PYTHON · PYTHON REGIUS

Keep your ball python the right way from day one.

A complete, fact-checked printable manual: enclosure and heat, the humidity range that decides everything, feeding and refusals, handling, health triage, the law in every state, and the routines that keep a ball python thriving for its full lifespan.

INSIDE

- ✓ Heat and humidity targets in Fahrenheit and Celsius, with a labeled enclosure diagram
- ✓ Prey sizing, thawing, and the reasons a ball python stops eating
- ✓ Health triage, a symptom table and an emergency card
- ✓ Owner logs, a pet-sitter sheet and a power-outage plan

46 pages

Beginner to Intermediate

Print or tablet

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Every page is written to be used, not just read once.

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START HERE

How to use this package

Three ways to get value from it, depending on where you are right now.

UNITS USED THROUGHOUT

Every temperature and size is given twice. Temperatures appear in degrees Fahrenheit (°F) with degrees Celsius (°C) beside them. Sizes appear in inches (in) and feet (ft), with centimeters (cm) beside them. A snake and its weight log are measured in grams (g), the unit a kitchen scale shows. Humidity is a percentage (%) of the moisture the air could hold. Prices are in United States dollars.

If you haven't set up yet

Check the law first, on page 23: a ball python is legal without a permit in most of the United States, but not everywhere. Then read pages 4 to 11 in order before buying anything. Enclosure material, heat control and humidity are the three decisions that are expensive to redo, and the three most often chosen wrong. Get them right the first time, and use the Budget & Shopping List on page 33 to buy once.

If your ball python is already home

Audit what you have against the Setup Checklist in the print list below, and go to the Red Flags page (page 24) if anything seems off. If the problem is a snake that won't eat, start on page 15 instead: refusal is common in this species and usually normal. If any heat source is running without a thermostat, fix that today, before anything else on this list.

Print these first

- | | |
|--|---|
| ☐ Page 31: Setup Checklist & Targets | ☐ Page 38: Pet-Sitter Sheet |
| ☐ Page 32: Emergency & Quick Targets Card | ☐ Page 39: Owner Log |
| ☐ Page 35: Symptom Quick Reference | ☐ Page 40: Equipment, Quarantine & Vet Log |
| ☐ Page 36: Daily, Weekly & Seasonal Routine | ☐ Page 41: Enrichment Checklist & Log |

WHY THIS PACKAGE EXISTS

Ball python keeping has moved a long way from the small-tub advice that dominated it for years. This package reflects current enclosure-size, humidity and enrichment standards, including the housing study on page 19, cross-checked against the veterinary and scientific sources listed on page 44.

NOT A SUBSTITUTE FOR

A reptile-experienced veterinarian. This is a husbandry reference for day-to-day care. Anything urgent or health-related is always a vet call, not a page in this book. Find a reptile vet before you need one and write the number on the emergency card. Nothing on the law page is legal advice either.

SECTION 01 · PROFILE

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Python regius</i> , called the royal python in Europe
Adult size	3 to 5 ft (90 to 150 cm); females significantly larger than males
Captive lifespan	20 to 30 years with good care, and some live far longer. About 10 years on average in the wild
Native range	Grassland, open forest and farmland across West and Central Africa, from Senegal to Uganda
Diet type	Carnivore: whole frozen-thawed mice and rats, one meal every 5 to 14 days depending on age (page 12)
Enclosure minimum (adult)	4×2×2 ft (120×60×60 cm, roughly 120 gallons), at least as long as the snake. PVC (polyvinyl chloride, a rigid plastic) preferred over glass
Humidity	55 to 65%, raised to 70 to 80% through a shed (page 9)
The one hard rule	Every heat source runs through a thermostat. No exceptions (page 8)
Handling difficulty	Beginner to intermediate. Defensive as a juvenile, generally mellow as a settled adult
Conservation	Wild populations are pressured by collection and export for the pet trade, and by habitat loss

Cost overview

Initial setup

\$40 to \$100 for a standard snake, \$220 to \$840 for the equipment, plus \$50 to \$150 for the first vet exam. Every line is on page 33.

Yearly ongoing

\$210 to \$550 a year, about \$18 to \$46 a month, for frozen rodents, substrate, electricity and the yearly vet exam.

BUDGET FOR THE WORST CASE TOO

A sick or emergency visit runs \$100 to \$500 or more, and imaging or surgery pushes that higher. Neither is likely if you follow this package, but an animal that lives 20 to 30 years will need a vet at some point. Set the money aside now rather than deciding in an emergency.

Fun facts worth knowing

- The name comes from the defensive pose, not the diet. A stressed ball python curls into a tight ball with its head tucked in the center, so a new snake that won't uncurl isn't being difficult. It just hasn't decided yet that hiding is unnecessary.
- It hunts with heat. Pits along the upper lip detect the body warmth of nearby prey, letting a ball python strike accurately in total darkness.
- A brooding female coils around her eggs without warming them. Brashears and DeNardo (2015) measured six python species at brooding temperatures and found no heat production in five, the ball python among them.

The species, morphs, size & lifespan

What you are buying, where it came from, how big it gets, and how long it stays.

One species

The ball python is *Python regius*, native to the grasslands, open forests and farmland of West and Central Africa. In the wild it eats small mammals, birds and amphibians, and it is an ambush predator that spends most of its life concealed, which is why cover matters so much in its enclosure. It is the most popular pet snake in the world.

Where pet ball pythons come from

Almost the entire wild supply traces to three countries at the center of the range: Benin, Ghana and Togo, which have exported close to four million ball pythons since 1978, most of them to the United States. Snakes were taken straight from the wild through the 1980s and 1990s; since the early 2000s most exports have been ranched, meaning eggs and gravid (egg-carrying) females are collected from the wild and the hatchlings raised briefly and shipped. The species makes up roughly 80 percent of all snake exports under Appendix II of CITES, the Convention on International Trade in Endangered Species of Wild Fauna and Flora, the treaty that tracks trade in species that could become threatened by it. The captive-bred hobby grew out of the 1990s, when the first albino ball pythons were bred.

Captive-bred snakes make the better pets: less likely to bite, more likely to accept the food offered, and less likely to carry parasites. Wild-caught and imported animals routinely carry much heavier parasite loads, and because the two get mixed at shows and in the trade, no seller's word replaces the fecal test on page 20.

Morphs, and what a price does not tell you

A morph is a selectively bred line with a different color or pattern. There are hundreds of them, combined into more than 7,000 recorded combinations. A standard snake runs \$40 to \$100, a common single-gene morph \$50 to \$150, and designer morphs \$250 to \$5,000 or more. One widely bred gene carries a known health cost: the spider morph is linked to wobble syndrome, a heritable neurological disorder that shows as side-to-side or twisting head movements, poor coordination, and trouble striking or constricting prey. Rose and Williams (2014) first described it from a survey of breeders and keepers, and Starck and colleagues (2022) found malformed balance organs in the inner ear of every spider morph they imaged. A spider, or any combination carrying the gene, may struggle to feed, and its offspring can inherit the condition. A wild-type or a single-gene morph with no known defect is the lower-risk first snake.

Size and lifespan

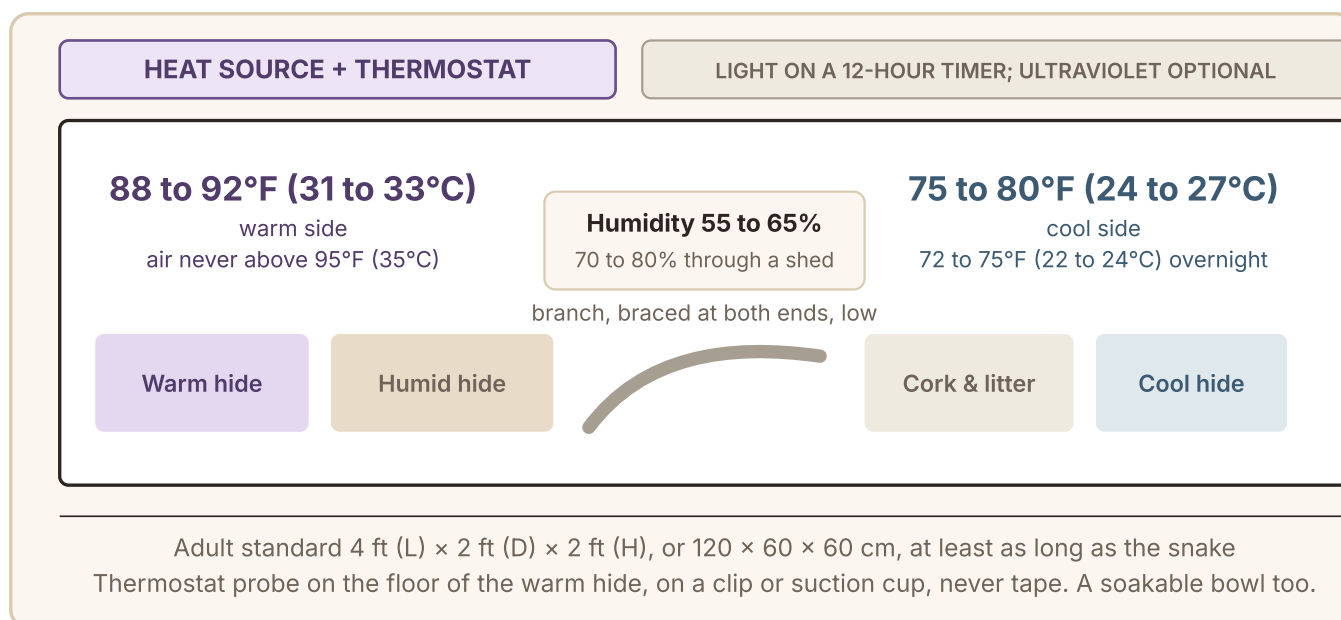
Growth	Hatchlings start at about a foot (30 cm) and reach roughly 3 ft (90 cm) by age three.
Adult length	3 to 5 ft (90 to 150 cm). Females are significantly larger than males.
Lifespan	20 to 30 years in captivity is common, against about 10 in the wild. Some live far longer: a zoo female laid eggs at 62 years old.

A ball python bought for a child can still be in the house when that child is grown. Plan for the whole span: the enclosure, the frozen rodents and a reptile vet for every one of those years.

Enclosure size, type & where it goes

4×2×2 ft (48×24×24 in, 120×60×60 cm, roughly 120 gallons and about 8 square feet of floor) is the adult standard, and the enclosure should be at least as long as the snake. Scale down while the snake is small: a hatchling up to about 300 g does well in a 10-gallon or 20×11×13 in (50×28×33 cm) enclosure, and a juvenile under 3 ft (90 cm) in a 36×18×18 in (90×45×45 cm) setup. Bigger, furnished enclosures have replaced the old advice that a small tub reduces stress.

PVC holds humidity far better than glass, which matters more for this species than for almost any other beginner snake. A glass tank works if you cover three sides to help it hold moisture. Whatever you use, it must be secure, locking and front-opening: ball pythons are excellent escape artists. House one snake per enclosure.



NEVER

Run any heat source without a thermostat. An unregulated mat, tape or panel is the direct cause of thermal burns and a real fire hazard. Skip hot rocks entirely.

Where it goes

- Out of the kitchen and away from anywhere food is made or served.
- In a room you can keep from dropping below the night range, 72 to 75°F (22 to 24°C), with the heat the enclosure has. Winter cooling is when respiratory infection spikes.

What the diagram doesn't show

- Cover across the whole floor, not just two hides. An enclosure that feels open causes chronic stress, seen as refused meals and skittishness.
- Hides sized snug, so the body touches the sides at several points. A hide the snake floats in gives no security.

The temperature gradient & heat sources

A warm end, a cool end, and a snake that moves between them to set its own body temperature.

ZONE	TARGET (°F)	TARGET (°C)
Warm side	88 to 92	31 to 33
Air temperature, anywhere in the enclosure	never above 95	never above 35
Cool side	75 to 80	24 to 27
Nighttime low	72 to 75	22 to 24

Check it with two tools, not the thermostat's own display: a digital probe thermometer left at each end for a continuous read, and an infrared temperature gun to spot-check the surfaces the snake lies on, the substrate, and the wall near the heat source. Run both checks before the snake goes in, and again whenever the setup changes.

Heat sources: what each one does

HEAT SOURCE	GOOD FOR	NOT GOOD FOR
Under-tank heat mat	Belly heat for a ground-dwelling snake; works through glass or thin PVC	Heating deep substrate or a large enclosure on its own
Ceramic heat emitter	A screw-in heater with no visible light, so it can run day or night	Fine-tuning low heat in a small enclosure
Deep heat projector	Like a ceramic emitter, with a more focused beam, for larger or taller enclosures	Small enclosures, where the warm spot can end up too small or too hot
Radiant heat panel	Even warmth over a broad area, mounted on or near the ceiling	One sharply defined warm spot
Halogen or incandescent bulb	Daytime heat with light, on a dimming thermostat	Night heat: light after dark disrupts the snake's day and night

Whichever you pick, wattage is a starting guess sized for an average room, not a target: wire it through the thermostat, then let the two measurements above settle it.

Heat that can't burn

- Keep any overhead heat source at least 4 to 6 in (10 to 15 cm) above the highest spot the snake can reach, behind a screened top or guard. A heater the snake can touch will burn it at any setting.
- For most pet snakes the warm end runs 90 to 95°F (32 to 35°C), so a surface reading well above that under the heat source is a fault, not a feature.
- Skip hot rocks entirely. No thermostat, direct skin contact and no cutoff is the textbook recipe for a thermal burn.
- A snake pressed against a hot surface can't always move away, especially while unwell, so sustained contact with an overheated spot is how burns happen.

Night heat has no light

A light bulb cannot be used for night heat. If the room falls below the 72 to 75°F (22 to 24°C) night range, use a heat mat, ceramic heat emitter or deep heat projector, on its own thermostat. None of them puts out visible light.

Thermostats & probes

The thermostat is the safety device

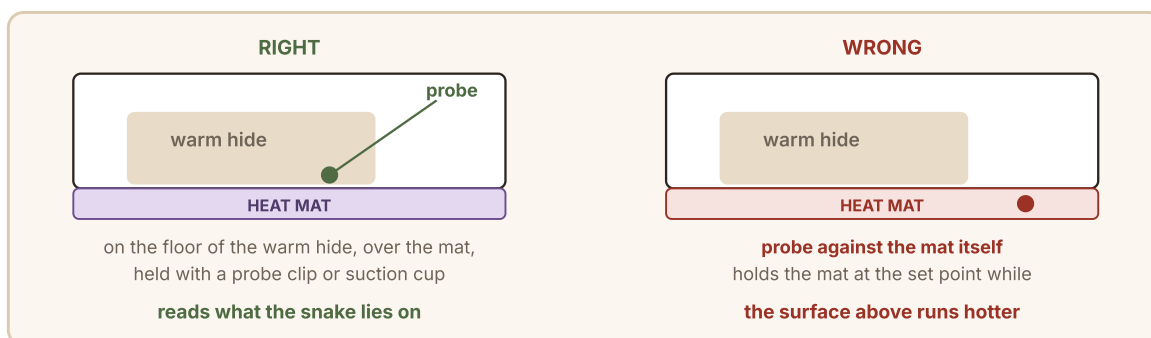
A thermostat measures temperature at one point and cuts power to the heat source once that point reaches the setting, then restores it as the reading drops. Without one, a mat or bulb draws full power with no ceiling at all. Thermal burns are a common and entirely preventable injury in this species, so budget the thermostat as part of the heat source, never as an upgrade, and never run a heat source unregulated, not even briefly while you wait for a part.

CONTROLLER	HOW IT WORKS	BEST MATCHED TO
On/off	Cuts power fully at the set point and restores it fully below it	Under-tank heat mats, which change temperature slowly enough for a full cycle
Pulse proportional	Sends short pulses of power to hold a steadier average	Ceramic heat emitters and other sources that react faster than a mat and can overshoot
Dimming	Turns the power down the way a light dimmer does	Halogen and incandescent bulbs, without the flicker of switching

The split is about how the device delivers power. What matters most is that every heat source has a controller at all.

Where the probe goes

A thermostat only regulates what its probe measures. For an under-tank mat, the probe goes on the floor of the warm hide, directly over the mat: the surface the snake actually lies on. Hold it there with a probe clip or suction cup. Tape over a probe throws off its reading, and a snake can stick to exposed adhesive and tear scales. A probe against the mat itself reads the equipment, not the snake, and lets the surface above run hotter than the display shows. For an overhead heater, follow your controller's own instructions; the principle is the same, the probe reads at the snake's level, never up near the fixture.



How a thermostat fails

- **The probe moves.** The most common failure, and a silent one: the display keeps showing an accurate reading for the wrong spot. Check the probe's position at every water change or spot clean.
- **The thermostat sticks on.** Rarer and more dangerous, because nothing limits the heat. Read the warm-end thermometer daily. The cool end always reads lower, and that gap is the gradient. A warm end that keeps climbing is the warning of a stuck thermostat or a shifted probe.
- **Test it before the snake arrives.** Run the whole setup and watch the thermostat cut the heat as the probe warms.

Humidity, light & optional ultraviolet

Humidity is where most ball python problems start, and it goes wrong in both directions.

HUMIDITY	TARGET
Everyday baseline	55 to 65%
Through a shed	70 to 80% through the shed cycle itself

Too dry, too wet

Too dry gives bad sheds and raises the risk of respiratory infection. Too wet without real ventilation swings the other way, toward scale rot and respiratory trouble from damp instead of cold. The species needs a stable middle, not a maximum. An enclosure can look perfect at a glance, with the right temperatures, clean substrate and a snake that seems fine, while quietly running too dry: humidity problems tend to show up weeks later as a bad shed rather than right away.

What holds it steady

- **A moisture-holding substrate**, 3 to 4 in (7.5 to 10 cm) deep.
- **An appropriately sized water bowl**, big enough for the snake to soak in.
- **A humid hide** packed with damp sphagnum moss, kept year round. A humid pocket the snake can enter and leave does a different job from a permanently damp enclosure.
- **PVC, or glass covered on three sides**. PVC holds humidity far better than glass.
- **Misting only when needed**, to bring a low reading back up.

Read it with a digital hygrometer (humidity gauge), not a cheap analog dial: dials are notoriously inaccurate.

Light: a timer, and dark nights

Run a 12-hour light, 12-hour dark day on a timer, using an LED (light-emitting diode, a low-heat, long-life lamp). Avoid bright white or colored bulbs at night. If the room needs night heat, it comes from a heater with no light (page 7).

UVB: optional, with a modest benefit

UVB (ultraviolet B) is the part of sunlight many reptiles need to make vitamin D3. Ball pythons have been kept successfully for decades with no UVB at all, so it is genuinely optional here, but a low-level gradient is a reasonable piece of enrichment if you want to add it. Ball pythons are partial-sun, occasional baskers (Ferguson Zone 2 in the grouping of species by natural sun exposure). That puts the UVI (ultraviolet index, the strength of the UVB at one spot) at 0.7 to 1.0 across the gradient and 1.1 to 3.0 at the closest point to the lamp, with a gradient the snake can move away from.

- Glass or plastic between lamp and snake blocks most UVB; a mesh lid costs some of it.
- A UVB tube keeps glowing long after its useful output has faded. Replace it on the maker's schedule, every 6 to 12 months depending on the model, and write the date down.

Substrate, hides & cover

Substrate

Cypress mulch, or coconut coir and husk, 3 to 4 in (7.5 to 10 cm) deep. Both hold humidity at the level this species needs. Paper towel works fine temporarily, for quarantine for example, where a bare floor is the point, but it won't hold humidity long term. Avoid aspen entirely: it tends to mold at ball python humidity.

NEVER

Pine or cedar. Both contain aromatic oils that are toxic to reptiles. This is a hard exclusion, not a preference, and it covers shavings, chips and bedding sold under any name.

Furnishings checklist

- ☐ At least two snug hides, one warm side and one cool side, sized so the body touches the sides at several points
- ☐ A humid hide packed with damp sphagnum moss, year round
- ☐ A heavy, tip-resistant water bowl the snake can fully get into
- ☐ Cork bark tubes and flats for cover across the floor
- ☐ Leaf litter and plants to break up open sightlines
- ☐ A sturdy branch braced firmly at both ends, low enough that a fall is not a fall

Why clutter matters for this species

Plants, branches and leaf litter aren't decoration. Ball pythons are naturally shy, and an enclosure that feels open and exposed causes chronic stress, which shows up as refused meals and general skittishness. The more useful reading of "give it a tight hide" is that the whole floor should offer cover, so the snake can be somewhere other than inside a box without being exposed. A cluttered enclosure gets used more, not less.

Climbing, which most setups skip

Ball pythons are heavy-bodied and not agile, so the requirement is stability, not height: a branch braced firmly enough that the snake can push against it without it shifting. Cork rounds stacked into a raised platform work as well and double as cover. Vertical structure adds usable space to an enclosure that otherwise only has a floor, and it gets used.

A HIDE THE SNAKE FLOATS IN IS NOT A HIDE

The most common furnishing mistake is buying hides that look comfortable to a person. A ball python feels secure when its body touches the sides at several points, so a snug hide does more for its stress level than a large decorative cave ever will. Replace hides as the snake outgrows them, and check the fit at every full substrate change.

Cleaning & hygiene

Cleaning removes the waste. Disinfecting kills what is left. Hygiene keeps it off your hands.

SALMONELLA

Healthy reptiles carry *Salmonella* bacteria in the gut and shed it in their droppings without being ill, on and off, so one negative test clears nothing. It travels from the snake, its enclosure and anything in it to hands, then to mouths or food, so wash with soap and running water after every contact. Do not kiss the snake or hold it to your face. Children under 5 should not handle the snake or touch anything in its enclosure, and adults over 65, pregnant people and anyone with a weakened immune system are also at higher risk.

Three clocks

WHEN	WHAT TO DO
As you see it	Snake urates (the white part of the waste) and feces soil a lot of substrate at once, so spot clean as soon as waste appears, scooping out the wet substrate around it.
Daily	Change the water, since the snake soaks in it.
Weekly	Scrub the water bowl and any soiled decor with a reptile habitat cleaner, or a 3% bleach solution (about half a cup of household bleach per gallon of water) left on for at least 10 minutes. Rinse until no smell remains, and let everything dry.
Full change	Replace the whole substrate monthly if the snake is messy, and at least once every 3 months even if it is not. Move the snake to a secure tub, strip the enclosure, disinfect every surface the same way, and let it dry completely before new substrate goes in.

Disinfectant only works on a surface already scrubbed clean, because a film of waste shields bacteria from it. Plan on a full day of airing after bleach before the snake goes back in.

NEVER THE KITCHEN SINK

Never wash the bowl, decor or enclosure in a kitchen sink, a bathroom sink or a bath people use, or anywhere food is prepared. Keep a plastic tub for the job, or work in a utility sink or outdoors. If a bathtub does get used, clean it thoroughly afterward. Wash bedding and decor separately from anything used for food, and keep the snake out of the kitchen entirely.

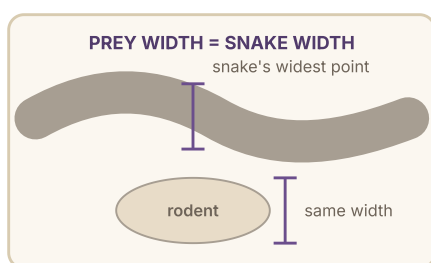
If someone in the house gets sick

The signs are ordinary ones: diarrhea, fever, stomach cramps. That is why the link gets missed. Tell the doctor there is a reptile in the house, even if nobody remembers touching it, since enclosure surfaces are enough on their own. It matters most for a child under 5, an adult over 65, someone pregnant, or someone with a weakened immune system, in whom the infection can spread beyond the gut.

Diet & feeding by age

Ball pythons are carnivores fed whole prey: frozen-thawed mice and rats, the nutritionally complete staple from hatchling to adult. Unlike most pets, the schedule gets slower as the animal gets bigger, not faster. A healthy adult needs a meal only every one to two weeks, and that slow metabolism is a feature of the species, not a problem to solve.

AGE	HOW OFTEN	WHAT
Hatchling, 0 to 6 months	Every 5 to 7 days	A frozen-thawed rodent as wide as the widest part of the body
Juvenile, 6 to 18 months	Every 7 to 10 days	The same width rule, moving up in size as the snake grows
Adult, 18 months and up	Every 10 to 14 days	The same rule; for most adults that is a medium rat



Prey size is the whole rule

Feed prey roughly as wide as the widest point of the snake's body. Too large risks regurgitation; too small gives too little nutrition over time. Resist the idea that a bigger snake is a healthier one: obesity is real and common in captive ball pythons, and a snake that eats every ten days is doing exactly what it should.

Where and how to feed

Feed inside the enclosure, with long tongs. Moving a ball python to a separate feeding tub was standard advice for years, but the added stress is now understood to raise both regurgitation and refusal in an already food-shy species. As long as loose substrate isn't a choking hazard, feeding in place is the better practice. Tongs every time also prevent the bites that come from a snake smelling rodent on your hand.

- **Wait 48 to 72 hours after a meal before handling.** Handling on a full stomach is one of the most common causes of regurgitation, which is hard on the digestive system and slow to recover from.
- **Water:** a heavy bowl big enough to soak in, changed daily, since the snake soaks in it as well as drinking from it.
- **Weigh it.** The scale catches overfeeding earlier than your eye does (page 16).

NEVER FEED LIVE PREY

A live rodent can bite and seriously injure a snake, sometimes fatally, especially one that isn't hungry enough to strike immediately. Frozen-thawed removes that risk entirely. The full never-feed list comes next.

Prey types & the never-feed list

Mice and rats are the staple, and for most snakes they stay that way for life. Everything else is a situational tool.

PREY	TIER	WHEN IT IS USED, AND THE CATCH
Frozen-thawed mice and rats	STAPLE	Feeder rodents bred for reptile feeding. Every regular feeding, from hatchling to adult. Nutritionally complete, with no reason to move off them long term.
African soft-furred rats (ASF)	OCCASIONAL	A different rodent species, milder-smelling and thinner-skinned. Used to restart a stubborn or newly arrived feeder, since the different scent can trigger a strike a snake is ignoring in a mouse or rat. A fix, not a diet: move back to mice or rats once the snake is feeding reliably.
Day-old chicks	OCCASIONAL	Usually sold frozen. Occasional variety, sometimes offered to a snake gone stubbornly off rodents. A different fat and nutrient profile, and more feathers and waste per meal. Not a staple.
Quail	OCCASIONAL	Whole quail or quail chicks. Variety feeding, used the same way as chicks for a stalled feeder, with the same feathers-and-waste tradeoff.

Rotating in more than one prey type is common, since nutrient content varies across prey species, but mice and rats alone are fine.

Never

NEVER	WHY
Live prey of any kind	A live rodent can bite and seriously injure the snake, sometimes fatally.
Prey wider than the snake's widest point	Causes regurgitation, which is hard on the digestive tract.
Prey thawed, then refrozen	Refreezing degrades quality and gives bacteria more chance to multiply in the second thaw. Thawed and uneaten is a loss, not leftovers.
Prey heated in a microwave	Uneven heating leaves hot spots that can burn the mouth while the inside stays frozen, and can rupture the prey.
Prey thawed on a counter for hours	Thaw in the refrigerator, or in a sealed bag in water, instead.
Prey fed cold, straight from the fridge	A cold rodent doesn't register as prey to the heat pits. Warm it gently first.
Overheated prey	Cooks it inside, risks the body cavity rupturing when the snake constricts or swallows, and can scald the mouth.

THE NEVER-FEED LIST, IN ONE PLACE

Live prey · prey wider than the snake's widest point · anything thawed and refrozen · anything heated in a microwave or left thawing on a counter · cold prey · overheated prey. Frozen feeder rodents can also carry *Salmonella* to the people who handle them, so the thawing steps come with safe-handling rules.

Thawing, warming & freezer storage

The step most often done carelessly, and it matters more than the schedule.

Thawing, step by step

1. **Thaw it fully.** In the refrigerator, the standard method, or sealed in a bag and submerged in cool to lukewarm water, changing the water as needed until there is no frozen core left.
2. **Never on the counter** for hours, and **never in a microwave**, which heats unevenly: the inside can stay frozen while pockets on the outside get hot enough to burn the mouth, and the prey can rupture.
3. **Warm it gently.** Don't feed it straight from the refrigerator. Prey goes in at room temperature or, better, a little warmer, never cold. A few minutes under a lamp, or warm water run over the outside of the sealed bag, does it.
4. **Judge by touch.** It should feel warm to the touch. Warm gradually and check often rather than leaving it under heat and walking away.

WHY WARMTH MATTERS

Ball pythons hunt partly by heat. The pits along the upper lip pick up infrared radiation from anything warmer than its surroundings, and a cold, thawed rodent doesn't register as prey the same way. Overheating is the opposite mistake: hot water or a lamp left too long can cook the prey inside, which risks the body cavity rupturing when the snake constricts or swallows, and can scald the inside of the snake's mouth.

Presenting it

Offer the warmed prey with long tongs, inside the enclosure (page 12). A snake that ignores a well-warmed meal is telling you something about its setup, its shed or the season, and the order to check those in comes next.

Freezer storage

- **Storage life.** Food held at a steady 0°F (–18°C) stays safe indefinitely, but quality declines the longer it sits. Don't let stock sit a year or more just because it is technically fine.
- **Freezer burn**, the gray, dried-out patches on long-frozen prey, is a quality problem rather than a safety one, and it happens when packaging isn't airtight. Keep prey in its original sealed packaging or rebag it well.
- **Thaw once.** Never refreeze a rodent that has been thawed. If you thaw more than the snake will eat, the extra is a loss.

HANDLING IT SAFELY YOURSELF

Frozen feeder rodents have been linked to several *Salmonella* outbreaks in people. Wash your hands with soap and water right after handling frozen rodents or anything in the area where they were stored, thawed or prepared. Keep them away from where human food and drink are stored, prepared, served or eaten, and use containers and utensils that are only for feeder rodents. Don't thaw prey in a sink you wash dishes in, and never use a microwave or kitchen utensil meant for human food.

Why a ball python stops eating

This species is famous for refusing food. A refusal in a snake in good condition is usually normal, not an emergency.

A healthy-looking ball python in good body condition that turns down a meal or two is not automatically in trouble. Check the causes in this order, the enclosure first:

	CAUSE	HOW YOU CAN TELL	DO
1	Temperature or humidity off	The most common husbandry cause. Read the thermometers and hygrometer, and check the thermostat probe	Fix the gradient and humidity before anything else (page 7)
2	A shed underway	Dull skin, cloudy or blue eyes	Wait. The refusal lifts on its own once the shed is complete (page 29)
3	Stress	A new enclosure, too much handling, or too little hide cover	Leave it alone, add cover, cut handling back
4	Breeding season	October to March, males especially	Completely normal. Weigh it and wait
5	The seasonal slowdown	The cooler months, with no change to the enclosure	Normal while its weight holds (page 22)

If everything checks out

With the temperature right, the humidity right and no other symptoms, it is often reasonable to offer food again in a couple of weeks before assuming anything is wrong. When you do, make the offer as good as it can be: the prey warmed properly, since a cold rodent doesn't register to the heat pits (page 14), and, for a stubborn or newly arrived snake, an African soft-furred rat, whose different scent can trigger a strike a mouse or rat does not (page 13).

How long is too long

A healthy adult ball python can go 3 to 6 months without eating and lose surprisingly little condition, and some fast through an entire breeding season without lasting harm. A female brooding eggs can go 4 to 6 months, living off fat built up beforehand. What turns a fast into a problem is what comes with it.

WHEN A REFUSAL BECOMES A VET VISIT

Rapid weight loss, or weight down more than 10% from the starting weight, or visibly thinner along the spine and tail base. **Lethargy** alongside the refusal. **Mucus around the mouth**, or **wheezing**. Still refusing by spring, or refusing out of season while losing weight. Any of these ends the wait.

REFUSED OR REGURGITATED?

A refused meal is a snake that never struck. A regurgitated meal was eaten and came back up. The first is usually normal and needs patience. The second means something in the setup, the handling or the snake's health is wrong, and it has its own plan (page 27).

Growth, weight & body condition

The scale says whether the feeding schedule is working. The cross-section says what that weight looks like.

Weight, not length

Length is easy to picture and the wrong number to build a growth record on, because length and body mass don't grow in step. Falk, Snow and Reed (2017), measuring 248 wild Burmese pythons, found mass climbs disproportionately faster than length as a snake gets bigger, so two snakes of the same length can carry very different muscle and fat. A gram or kitchen scale captures muscle, fat and gut content in one number.

- **Monthly** is a reasonable default for a growing juvenile, less often once an adult has leveled off.
- **Every 1 to 2 weeks** through a long refusal or the winter slowdown, and **weekly** during quarantine.
- Same scale every time, and write every reading on page 39. One weigh-in tells you almost nothing; three or four that climb, hold flat or drop tell you a great deal.

What growth looks like

Hatchlings start at about a foot (30 cm) and reach roughly 3 ft (90 cm) by age three; adults settle at 3 to 5 ft (90 to 150 cm), females significantly larger. No verified weight ranges by age exist for snakes in any veterinary or peer-reviewed source, so this package prints none. That is fine: a steady climb in a juvenile and a stable line in an adult are what matter, whatever the numbers.

Body condition: the cross-section test

Pick the snake up, support it well, and look at it end-on, imagining a slice straight across the body. Run a gentle hand along the back.

UNDERWEIGHT

Triangular in cross-section, with the muscle sloping away from a visible ridge down the center of the back, where the spine sits closer to the surface.

IDEAL

A rounded loaf or slightly rounded rectangle, full and even with no sharp angles. Along the back it feels like muscle over bone, not a sharp ridge or a padded gap.

OVERWEIGHT

Round in a different way: skin folds along the sides, or a soft, doughy feel over the ribs where fat has built up under the skin.

The 9-point body condition scale used for dogs and cats does not apply to snakes. Use this check at every weigh-in.

WHEN A BODY CHANGE IS A VET VISIT

A snake that reads triangular over several checks, especially with a falling weight line, refused meals or lethargy, needs a vet rather than more food: thinness in a reptile is as often parasites, illness or a husbandry problem as it is underfeeding. A snake that is consistently round needs its prey size or feeding frequency adjusted, never meals skipped outright, since sudden fasting brings its own problems in an animal already carrying extra weight.

Handling & taming

Defensive as a baby, mellow as an adult. Most of what makes handling go well is timing, not grip.

Give it time to settle first

Don't handle a newly arrived ball python for the first one to two weeks, and don't start until it is eating regularly on a normal schedule. A snake still adjusting to a new enclosure, with no feeding rhythm yet, is not ready to be picked up. Until then: fresh water, spot cleaning, and nothing else. Juveniles are often more defensive and high-strung than adults; that is close to universal, and it almost always settles as the snake matures. Patience early pays off later.

How often and how long

Once the snake is settled, 2 to 3 sessions a week is a reasonable baseline, up to 3 to 5 for a snake that handles well, keeping each one to 15 to 30 minutes at most. Overhandling causes real stress in this species and is a common cause of refused meals, so more is not better here.

The two timing rules

48 to 72 hours after a meal

Handling too soon after feeding is one of the most common causes of regurgitation in pet snakes, and a regurgitation is hard on the digestive system and slow to recover from.

Never during a shed

Cloudy or blue eyes mean a shed has started and the snake is functionally blind for a few days. A snake that can't see is more defensive and more likely to bite out of startlement, not aggression. Wait until the shed is complete and the eyes are clear.

How to pick one up

Approach from the side, never straight down from above, which reads as a predator. Support the snake's full weight with both hands or along your forearms, lifting from the middle of the body rather than near the head or tail. Let it move across your hands at its own pace with gentle, steady control instead of restraining it. Check the enclosure lock after every session.

PUT IT BACK WHEN YOU SEE

Balling up tightly with the head tucked away · rapid or heavy breathing · the head hidden against your hand or arm · actively trying to move away · musking, a foul-smelling release from the cloaca (the single opening under the tail). None of these is aggression, and a refused next meal is the same message, late.

Bites

Rare, and usually not aggression. A bite is typically a feeding response to the smell of rodent on your hands, or a startled reaction. The wound is generally a small skin break, cleaned with soap and water. Feed with tongs, and don't handle the snake right after you have handled rodents.

Who should hold it

Supervise children around the snake at all times. Children under 5 should not handle it at all, because a healthy-looking snake can still carry *Salmonella* (page 11). Wash hands with soap before and after every session.

Body language, normal behavior & escapes

What a ball python is telling you, which signals are normal, and which are a job for you or a vet.

YOU SEE	IT USUALLY MEANS	DO
Coiled into a tight ball, head in the center	Fear, the defensive pose the species is named for, whether the trigger is real danger or mild stress	Treat it as a request for space
Hiding most of the day	Normal: an ambush predator spends most of its life concealed	Nothing. Make sure there is cover across the floor
Musking during handling	It feels seriously threatened	End the session
Dull skin, eyes cloudy blue	A shed coming. The eyes stay cloudy about 4 to 7 days	No handling; raise humidity through the shed
Eyes clear again, skin still dull	Not finished: the shed usually follows about 4 to 7 days later	Keep waiting, and keep hands off
Soaking in the bowl for long stretches	Can be a sign of mites, which snakes seem to try to drown	Check closely around the eyes and chin (page 26)
Off food from fall into winter	The seasonal appetite dip, roughly October to March	Normal while weight holds (page 22)
Wheezing, clicking, open-mouth breathing	Respiratory infection	Vet promptly (page 25)
Head held up and back, tremors, can't right itself when turned over	"Stargazing" and other nerve signs, the mark of inclusion body disease (IBD), a fatal virus of boas and pythons	An emergency, not a behavior (page 28)

If it gets out

Ball pythons are skilled escape artists, and a snake can stay alive loose in a house for a long while, even if the search takes a week. Search in this order:

1. **Secure other pets.** Cats and dogs shut away, especially overnight, when the snake is most likely to move.
2. **Start at the enclosure.** Behind, under and around it first, then outward along walls and baseboards: snakes travel along edges, not across open floor.
3. **Clear one room at a time.** When a room is done, close the door and push a towel into the gap underneath.
4. **Check warm, tight places:** in and under couch cushions, beds and box springs, bookcases and dressers, and behind and under the fridge, stove and dishwasher. Check the oven before turning it on.
5. **Search at night with a flashlight,** when a loose snake is most likely moving.
6. **Leave the enclosure open with its heat on.** Some snakes return on their own. A shoebox with a snake-sized hole and a shipping heat pack inside, in a room kept around 68°F (20°C), gives it a warm hide to choose. Crumpled plastic bags along the baseboards rustle when a snake crosses them.

Once it is back, check it over: a snake loose for days may be dehydrated, hungry or injured. Look for cuts and stuck debris, offer water, feed it once it has settled if it missed a meal, and watch for wheezing or mouth discharge over the following weeks. Then find the gap it used and close it.

Enrichment & common mistakes

Mistakes that cause the most harm

- Running a heat source without a thermostat, or putting the probe against the mat instead of on the surface the snake lies on.
- Treating the small tub as the goal rather than the compromise, and never moving the snake into a real enclosure.
- Guessing at humidity, or trusting an analog dial, instead of reading a digital hygrometer.
- Leaving the enclosure bare because clutter looks untidy, in an animal whose stress is driven by feeling exposed.
- Feeding in a separate tub, or handling within 48 to 72 hours of a meal, then treating the regurgitation that follows as bad luck.
- Feeding an adult on a juvenile's schedule. Obesity is real and common in captive ball pythons.
- Skipping quarantine for a new snake, which is how mites and inclusion body disease reach a collection (page 20).
- Pine, cedar or aspen substrate, or live prey.

Enrichment, by priority

Enrichment comes after a thermal gradient, correct humidity and secure hides, never instead of them. For this species the key question has been tested: Kaufmann and colleagues compared 35 ball pythons in a conventional rack with furnished terrariums (substrate, hides, climbing, a basin big enough to bathe in, raised basking spots, living plants) and concluded that for welfare, racks are no alternative to a furnished terrarium. They compared whole systems, not single items. In this order:

1. A furnished enclosure

The part with a study behind it. Treat the enclosure itself as the enrichment, not something you add items to.

2. Cover across the whole floor

Not just two hides. Cork tubes and flats, leaf litter and plants let the snake be out without being exposed. Two secure hides, warm and cool, stay the baseline.

3. Stable climbing structure

A branch braced at both ends, or stacked cork, low enough that a fall is not a fall. Racks cannot offer this at all.

4. Water big enough to get into

A basin the snake can fully enter supports shedding and hydration, and it is passive: you provide it, the snake decides. Keep it stable and clean, since a soaking snake also defecates in it.

5. Scent

Snakes navigate chemically. A shed skin from another enclosure, a scent-marked piece of cork, or prey dragged across the substrate before offering costs nothing. This is reasoning from natural history, with no study behind it.

6. Rearranging, genuinely last

A new piece of cork or a shifted branch every few weeks is fine. Tearing the enclosure down weekly is not.

WHAT NOT TO DO

Don't treat a rack as enriched by adding one item to it: the rack is the thing that lost. Don't stack decor so densely the snake can't reach the warm spot; thermal access outranks everything here. Don't use handling as enrichment, especially around a shed. And don't let feeding response drive the design: a snake that strikes at movement is not asking for a more exciting room.

Choosing a snake, quarantine & the first vet visit

Captive-bred, already eating, checked over in good light, and in front of a vet within 72 hours.

A breeder who hatched the snake can tell you when it hatched, what it eats, and when it last ate and shed. Buy only where the enclosures are clean and large enough. Ask to see the snake eat, or for its feeding record: a ball python already taking frozen-thawed rodents on a regular schedule is worth more than one that is not. A rescue or an expo animal often comes with no reliable history, which is not a reason to refuse it but a reason to run the full quarantine below.

Look before you pay

CHECK	HEALTHY	WALK AWAY OR ASK MORE
Eyes	Clear and alert	Swollen, sunken, crusted, or cloudy outside a shed
Nose, breathing	Dry nostrils, quiet breathing	Bubbles, mucus, wheezing, open-mouth breathing
Vent	Clean	Stained, pasted with feces, or swollen
Body	Rounded, no projecting spine or ribs	Visible bones, or well behind its age in size
Skin and shed	Intact, no old shed left on	Wounds, scratches, or rings of old shed on the tail tip
Mites	None visible	Tiny black, brown, reddish or orange flecks, especially around the eyes
Behavior	Tongue flicking, aware, good muscle tone	Limp or unresponsive

Ask the seller: the hatch date; whether it was captive-bred; what it eats and when it last ate; and whether it has been treated for anything, mites or a respiratory infection above all. A good breeder answers all four without looking anything up.

Quarantine: 3 to 6 months

If another reptile already lives in the house, keep the new snake fully apart for 3 to 6 months. 60 to 90 days is a floor, not a target: mites usually show within weeks, but IBD can sit silent for months to years. With no other reptile, skip the separation but not the vet visit.

- **A separate room.** Two enclosures on one shelf share the same air.
- **Paper towel, a hide and a water dish.** Paper shows mites, stool and regurgitated food at a glance.
- **Separate tools, and the new snake last.** Its own tongs, bowl and scraper, cared for after every other animal, hands washed in between.
- **Weigh it weekly, and restart the clock** if anything flags: a positive fecal test, a regurgitated meal, unexplained weight loss. At the end, disinfect everything that touched it.

THE FIRST VET VISIT, WITHIN 72 HOURS

Book a reptile vet before the snake comes home (finding one is on page 24), for within 72 hours of arrival and no later than two weeks. The exam records weight, general condition and mobility, and the vet checks a fresh stool sample for parasites. Bring one in a sealed bag. Treat that fecal test as non-negotiable: parasites often show nothing for months. The visit is the baseline every later one is measured against.

Telling the sex, females & eggs

Not a do-it-yourself job, and worth knowing whether or not a male has ever been in the house.

Telling male from female

Ball pythons show almost no external sex difference before adulthood, and adult females are simply larger. Confirming sex means checking the internal anatomy, by one of three methods. Probing and popping are easy to describe and hard to do right on a live animal: the wrong pressure, angle or probe size can injure the reproductive organs or the tail. Ask a reptile-experienced vet to do it at a routine visit, or have an experienced breeder walk you through it in person first.

METHOD	HOW IT WORKS	USED ON	WHO
Probing	A lubricated, blunt probe slides into the cloaca toward the tail. In a male it reaches 6 to 14 scales deep, into the pocket that holds a hemipenis (one of the paired male organs); in a female it stops at about 3 to 6	Adults and subadults	Vet or experienced breeder only
Popping	Gentle, controlled pressure at the base of the tail turns out a male's hemipenes so they can be seen	Hatchlings and very young snakes	Vet or experienced breeder only
Ultrasound	Imaging of the internal reproductive organs. Reliable in snakes, with no injury risk	Any age	Reptile-experienced vet

Read nothing into cloacal spurs, the small claw-like points either side of the vent. They are a real sex-linked trait in boas and pythons, and Hoefer, Robinson and Pinou (2021) found male Bahamian boas carry spurs nearly twice as long as females, but that is one wild boa population measured as a group. It gives you no cutoff for the single snake in your hands. Tail shape is a weaker guess still.

Females and eggs

Ball pythons lay eggs. A female lays a clutch and coils around it, and Brashears and DeNardo (2015) found she does not warm the eggs by shivering, as pythons are often credited with doing: they measured no heat production in the ball python. A brooding female can go 4 to 6 months without eating, living off fat built up beforehand, so a long fast in a female who has laid is expected.

A female kept alone can still produce eggs. A zoo female laid seven eggs at 62 years old, more than 15 years after her last contact with a male, either from long-stored sperm or from facultative parthenogenesis, reproduction without a male.

A SAME-DAY VET VISIT

Straining with nothing passed, or any tissue protruding from the vent. Egg binding, a female unable to pass her eggs, is one of the causes of straining hard enough to push tissue out of the vent, and a prolapse is always an emergency (page 28).

The winter appetite dip & brumation

A seasonal slowdown, not a medical crisis, once you have checked it is not illness.

Brumation is the reptile version of a seasonal slowdown, tied to temperature and daylight: less activity, less appetite, a slower metabolism. It is not mammal hibernation. A brumating snake stays capable of waking, moving and surfacing to drink. Many ball pythons show a natural dip in appetite from roughly October through March with no deliberate cooling at all, and they normally start eating again on their own in late winter or early spring. Wild ball pythons live with modest seasonal temperature swings, which is one reason the captive version usually looks like a mild appetite dip rather than a dramatic shutdown.

Seasonal slowdown or illness?

Illness also starts with a refused meal, so the question is never the appetite alone. Check the rest of the snake:

CHECK	NORMAL SEASONAL SLOWDOWN	ILLNESS: CALL A REPTILE VET
Enclosure temperatures	Unchanged, checked with a thermometer at both ends	A failed heater or thermostat explains the fast: fix it first, then watch
Weight, every 1 to 2 weeks	Stable, or a slow drift well inside 10% of the starting weight	Down more than 10%, or visibly thinner along the spine and tail base
Breathing	Quiet, mouth closed	Open-mouth breathing, bubbles or discharge at the nose or mouth, or wheezing
Mouth	Pink, clean gums	Excess saliva, tiny red spots on the gums, or cheesy yellow material along the teeth
Behavior	Hides more and prefers the cool end, but moves normally when handled	Lethargy well beyond the usual slowdown
Timing	Starts in fall, ends by spring	Still refusing by spring, or refusing out of season

A snake that passes every line can be left alone with water available and weighed every one to two weeks. Any line in the right-hand column is a reason to book a vet visit now. Respiratory infections get worse in a snake kept too cool, which is why a sick snake is kept at the upper end of its normal range, never cooled.

Does a pet ball python need brumating?

No. Deliberate cooling is a breeder's tool for triggering a breeding cycle, and a snake kept as a pet at steady, correct temperatures year round doesn't need it. Breeders cool only healthy snakes of a good weight with a reptile vet's clearance: feeding stops 1 to 2 weeks before so the gut is empty, temperatures come down gradually over 1 to 2 weeks, the cool period typically runs 60 to 90 days with water always available, and the snake is weighed before and throughout. Weight loss beyond the expected range, any breathing sign, or lethargy well beyond the season's quiet ends it: warm the enclosure back up. Warming happens as gradually as cooling, and the first meal waits until the snake has been back at normal temperatures for several days and is active and alert.

Is a ball python legal where you live?

Legal without a permit almost everywhere in the United States, with a short list of exceptions.

This page covers all 50 states, the District of Columbia and New York City, which has its own rules: 52 jurisdictions. A pet ball python is **legal with no permit in 46, barred in 2, permit-only in 3, and legal on a condition in 1.**

WHERE	STATUS	THE RULE
Hawaii	Barred	Every snake species is prohibited: "Serpentes (all species)" on the List of Prohibited Animals (Hawaii Administrative Rules § 4-71-6).
New York City	Barred	The city Health Code bans "any member of the family Pythonidae" (§ 161.01(b)(10)), which takes in the ball python although it is never named. New York State itself allows it.
New Jersey	Permit	The permit list names "Pythons, Family Pythonidae" (New Jersey Administrative Code 7:25-4.3(a)3.i), so a ball python needs the individual hobby permit.
Delaware	Permit	A non-native reptile on no exempt list, so it needs a \$25 Individual permit from the Department of Agriculture (3 Delaware Administrative Code 903).
West Virginia	Permit	West Virginia Code § 20-2-13 bars bringing in or keeping live wildlife from out of state without an importation permit from the director, captive-bred or not. No application is published: ask the state in writing first, and keep the reply.
Minnesota	Conditional	Minnesota Statutes § 97A.401, subdivision 8: a snake from a permitted breeder, or held before 1 August 2021, may be kept as a pet. The breeder permit is not visibly in place yet, so ask the state wildlife agency how to meet the condition.

Legal with no permit: 46 jurisdictions

Alabama, Alaska, Arizona, Arkansas, California, Colorado, Connecticut, District of Columbia, Florida, Georgia, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maine, Maryland, Massachusetts, Michigan, Mississippi, Missouri, Montana, Nebraska, Nevada, New Hampshire, New Mexico, New York State, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, South Dakota, Tennessee, Texas, Utah, Vermont, Virginia, Washington, Wisconsin and Wyoming.

In twelve of them, California, Colorado, Illinois, Kentucky, Maryland, Michigan, Montana, Nevada, North Carolina, Ohio, Oregon and Washington, the state lets cities or counties be stricter, so check your local ordinance (a city or county law) too. Illinois also requires a strong, impact-resistant, escape-proof enclosure for any non-native python. Bringing one into Michigan calls for a certificate of veterinary inspection, into Wisconsin an animal import permit, and into South Dakota a free entry permit.

THE FEDERAL PYTHON BAN NEVER INCLUDED IT

The 2012 federal rule under the Lacey Act listing invasive constrictors as injurious wildlife names four species: the Burmese python, the Northern and Southern African pythons and the yellow anaconda. The ball python was considered and left off, and the Florida and Georgia lists stop at the same large invasive species.

NOT LEGAL ADVICE

State legal status as of October 2026. This is a summary of state rules, read on 1 October 2026 (Maine on 5 August 2026), and not legal advice. Rules change. Before you buy or move, check with your state wildlife agency and with your city or county.

Red flags & finding a vet

What to watch for, what to tell the vet, and how to find one before you need one.

CONTACT A REPTILE VET PROMPTLY IF YOU SEE

Wheezing, clicking, open-mouth breathing, or mucus or bubbles at the mouth or nostrils (urgent) · belly scales discolored, spreading or blistering · pinpoint bleeding on the gums, pus, cheesy material or a sour smell in the mouth · visible mites · any tissue protruding from the vent · the head held up and back, tremors, or a snake that can't right itself · a burn · a meal brought back up · a refusal that comes with weight loss, lethargy, mucus or wheezing · watery stool more than once in a row, blood in the stool, or straining with nothing passed · an eye cap left on after a shed.

What to tell the vet

- ☐ Warm side and cool side temperatures, and whether every heat source runs on a thermostat
- ☐ Current humidity, read on a digital hygrometer
- ☐ Date of the last meal, its size, and whether it stayed down
- ☐ When symptoms started and whether they're worsening
- ☐ Last bowel movement, and what it looked like
- ☐ Weight trend from the Owner Log, and a fresh stool sample in a sealed bag if you can

Why temperature and humidity come up first

Ball pythons are generally hardy, but sensitive to husbandry mistakes in specific, predictable ways. Most of what shows up in this section traces back to temperature, humidity or hygiene, which is why a vet asks about your setup before anything else, and why most of it is preventable.

Finding a reptile vet

Not every practice treats reptiles, and the time to learn that is not during an emergency. The Association of Reptile and Amphibian Veterinarians runs a find-a-vet directory of its members, and at minimum your vet should be a member in good standing. When you call a clinic, ask how many reptiles it sees, whether it treats ball pythons routinely, whether it runs fecal parasite tests in house, and who covers reptile emergencies after hours. Write the number on page 32. After the first exam, a ball python should see the vet once a year.

Normal or concerning

NORMAL	CONCERNING
Refusing a meal or two in good body condition, especially from fall to spring or during a shed	Refusal with weight loss, lethargy, mucus at the mouth or wheezing
Balling up when startled, then relaxing once left alone	Lethargy well beyond the usual winter quiet
Dull skin and cloudy blue eyes for several days before a shed	Shed coming off in pieces, or eye caps still on afterward
Hiding most of the day, and soaking now and then	Soaking for long stretches, or rubbing against surfaces, with specks around the eyes
Stool with loose fur and bone fragments, a white urate	Yellow, orange or gritty urates, watery or bloody stool

Respiratory infection & scale rot

One from too cold or the wrong humidity, the other from too wet and too dirty.

Respiratory infection

The signature ball python illness, and the one every keeper should be able to recognize. **Cause:** temperatures running too low, humidity wrong in either direction, a dirty enclosure, or ongoing stress. Vets see a seasonal spike each winter as enclosures cool along with the weather outside, and a power outage that holds the snake below 72°F (22°C) for more than a day or two is the classic trigger. **Signs:** wheezing, an audible clicking, open-mouth breathing, mucus or bubbles around the mouth or nostrils, nasal discharge, lethargy, and sometimes gaping.

Response: it needs antibiotics prescribed by a vet, not a wait-and-see approach. Left untreated it can progress to pneumonia and become fatal. While you arrange the visit, raise the enclosure toward the top of its normal range, check the humidity, and isolate the snake if you keep others. A sick reptile is kept at the upper end of its preferred range because warmth there helps the immune system and the drugs work, which is also why an enclosure shouldn't sit at the bottom of its range by default.

LEARN THIS ONE BY SOUND

Clicking or wheezing is the clearest audible signal in this package. If you hear it: warm side to target, humidity checked, the snake isolated from others, and a call to a reptile vet today.

Scale rot (infectious dermatitis)

Cause: damp, dirty substrate, excess moisture, or poor hygiene in the enclosure. It is the exact failure of chasing a humidity number without ventilation. **Signs:** discolored belly scales, brown, reddish or yellow; blisters or sores; scales that feel soft or damaged; and a foul smell.

Response: move the snake to a clean, dry enclosure immediately and correct the humidity. Mild cases respond to antiseptic or antibiotic treatment, but anything spreading or blistering needs a vet.

CONCRETE THRESHOLD

Any belly scale that is discolored and spreading, or blistered, is a vet visit, not a wait-and-watch. The fix that prevents both conditions is the same: humidity held in its band with real ventilation (page 9), and the cleaning schedule on page 11.

Mouth rot, mites & thermal burns

One rides on another problem, one arrives on a new animal, and one is prevented by a single device.

Mouth rot (infectious stomatitis)

Often secondary to stress, injury or ongoing poor husbandry rather than arriving on its own. **Signs:** pinpoint bleeding on the gums, thick excess mucus or pus, cheesy-looking buildup in the mouth, a sour smell, visible swelling, open-mouth breathing, and refusing food. **Response:** it needs veterinary treatment and does not resolve on its own. Because it usually rides on another problem, bring an honest account of your setup to the visit.

Mites

Snake mites (*Ophionyssus natricis*) are tiny blood-feeding parasites that spread fast through direct contact and shared equipment, which is how a new animal brings them in. **Signs:** tiny black, brown, reddish or orange specks on the scales, clustered around the eyes, the chin or under the scales; long soaking sessions, as if trying to drown them; rubbing against surfaces; a drop in appetite. A heavy burden kills through anemia or infection, and mites carry bacteria and viruses between snakes, inclusion body disease among them.

- **Check:** gently rub the snake while it rests over a sheet of white paper, and look for anything that falls off.
- **Treat:** see a vet once mites are confirmed. Treating the snake alone is not enough: the whole enclosure has to be deep-cleaned at the same time, because mites persist in the environment after the snake looks clear.

Thermal burns

A common and entirely preventable injury, caused by a heat source running without a thermostat, a probe that has slipped off the surface it should read, or a heater the snake can touch. Hot rocks are the textbook case. Every heat element runs through a thermostat, with its probe where the snake lies (page 8) and any overhead heater at least 4 to 6 in (10 to 15 cm) out of reach. A burn is a wound: fix the heat source first, then see a reptile vet.

QUARANTINE IS THE REAL FIX FOR MITES

A new snake kept apart for 3 to 6 months, on paper towel, with its own tools and cared for last, is how mites are kept out of a collection in the first place (page 20). On bare paper, checked daily, mites usually show within the first few weeks if they are there at all.

Internal parasites, regurgitation & obesity

Less dramatic than the last two pages, and far more common.

Internal parasites

Every newly arrived ball python should get a fecal exam before you assume it is healthy, even if it looks fine and eats well. Parasites often cause no visible signs for months, and by the time they do, the burden can already be significant. Wild-caught and imported animals carry far heavier loads than captive-bred snakes, and the two get mixed at shows and in the trade.

A FECAL TEST FINDS	WHAT IT MEANS
Protozoa (single-celled parasites)	Most found in reptile stool are harmless. Some, such as <i>Entamoeba invadens</i> , cause serious illness in snakes.
Nematodes (roundworms)	<i>Strongyloides</i> and <i>Rhabdias</i> turn up often, ranging from minor to a cause of gut blockage or pneumonia, depending on the species and the load.
<i>Cryptosporidium serpentis</i>	The one that worries vets most. It thickens the stomach lining and causes repeated regurgitation after meals, has no reliably effective treatment, and can be fatal.

Watch for: weight loss in a snake still eating normally, regurgitated meals, and abnormal stool: watery, heavy with mucus, or with visible worms. **Treatment** depends entirely on what the test finds. The vet picks the drug and the dose for the parasite identified; this is never a guess or an over-the-counter product.

Regurgitation

A meal eaten and then brought back up. It is hard on the digestive system and slow to recover from. The usual causes are handling within 48 to 72 hours of a meal, prey wider than the snake, the stress of a separate feeding tub, or an enclosure at the wrong temperature. It can also be a parasite sign, and in pythons an early sign of inclusion body disease.

AFTER A REGURGITATION

Stop feeding for a cycle and call a reptile vet, especially in a python. Fix any of the causes above before the next meal.

Obesity

Real and common in captive ball pythons, and the result of overfeeding. **Signs:** a body round in cross-section, with skin folds along the sides or a soft, doughy feel over the ribs (page 16). **Fix:** adjust the prey size or the feeding interval to the schedule for its age, never by skipping meals outright, since sudden fasting brings its own problems. The scale catches it earlier than your eye does.

Inclusion body disease & prolapse

One has no cure and is kept out by quarantine. The other is always an emergency.

Inclusion body disease (IBD)

A fatal viral disease of boas and pythons, caused by reptarenaviruses. It shows as nerve signs, most recognizably **stargazing**, the head held up and back at an odd angle as if staring at the sky. Other signs: tremors, facial tics, abnormal tongue flicking, disorientation, loss of the righting reflex (the snake can't turn itself back over when flipped), regurgitation, and in advanced cases seizures.

How it spreads	Through body fluids: bites during breeding, fight wounds, contaminated droppings, and snake mites carrying infected blood between animals.
Boas and pythons differ	A boa can carry the virus for months to years with few or no signs. A python goes the other way: once signs start, most die within days or weeks.
Diagnosis	PCR (polymerase chain reaction, a lab test that finds a germ's genetic material) and a biopsy.
Treatment	None. Once nerve signs appear, most vets discuss euthanasia rather than treatment.
Other reptiles	Not established in corn snakes and other colubrids, or in lizards, so treat it as a boa and python risk.

QUARANTINE IS THE ONLY DEFENSE

A carrier can look completely healthy for the whole of a short quarantine, which is why a real one runs 3 to 6 months, in a separate room, with the new snake cared for last (page 20). Any nerve sign in a new or established snake is an emergency: keep it well away from every other reptile and call the vet.

Prolapse

Internal tissue pushed out through the vent and visible from outside. Most often it is the cloaca itself, but it can be the hemipenes in a male or a section of the colon. **It is always an emergency.**

- **Keep the tissue clean and moist** with a damp paper towel or a clean cloth, and get to a vet immediately. Exposed tissue dries and swells quickly, and the longer it stays out, the harder it is to save.
- **Never try to push it back in** yourself.
- **It is a symptom, not a disease.** Straining from constipation or a blockage, dehydration, internal parasites, egg binding or bladder stones can all push tissue out. The visit is about finding and fixing that cause too, or it tends to recur.

Shedding, stuck shed & eye caps

A healthy shed comes off in one piece. Look at every one.

The shed cycle

STAGE	WHAT YOU SEE
Pre-shed	Colors dull, eyes turn cloudy blue, the snake hides more and may refuse food. The eyes usually stay cloudy about 4 to 7 days.
Clearing	The eyes go clear again. The snake is not finished: the shed usually follows about 4 to 7 days later.
Active shed	The snake rubs against rough surfaces to start the peel, then crawls out of its skin.

Humidity doesn't separate the old skin from the new: a fluid the snake releases under the old skin, carrying enzymes, does that, and the cloudy blue eye is that fluid seen through the clear scale over the eye. Humidity matters at the end, when the loosened skin has to come away in one piece. So through a shed, raise humidity to 70 to 80%, make sure the humid hide is genuinely damp, give the snake cork, slate or a rough branch to rub on, and leave it alone: no handling.

Stuck shed (dysecdysis)

Skin that comes off in flakes and patches instead of one piece. Low humidity is the most common single cause, but one of six: skin parasites, poor nutrition, infection, nothing rough to rub against, and low thyroid function all do it too. Rings of old skin left on the tail tip can tighten across shed cycles and cut off its blood supply.

- **To loosen it**, soak the snake in chin-deep water at enclosure temperature for 30 minutes, supervised throughout, then gently work the loose skin free. Never peel unshed skin by force.
- **Repeated bad sheds in a correct setup** point to parasites, nutrition or an underlying condition. That is a vet conversation, not a humidity problem.

Retained eye caps

The clear scale over each eye, the spectacle or eye cap, sheds with the rest of the skin. When it stays on, the snake's vision and behavior often look completely normal, which makes it one of the easiest problems to miss. After every shed, find the skin and check that both eye caps came off with it.

NEVER PICK AT AN EYE CAP

Removing a retained eye cap is a veterinary job: days of eye ointment until the cap falls off or can be lifted with fine forceps. The eye underneath is not replaceable. See a vet rather than reaching for tweezers.

Reading droppings, urates & hydration

Stool is the earliest free health check you get. Look at it before you clean it up.

Urates are the white part of a reptile's droppings. Instead of liquid urine, the body turns nitrogen waste into uric acid, made in the liver from protein digestion, and passes it as a white, chalky paste. A snake has a single opening for both, so the urate comes out with the stool. Its color and texture are the fastest daily read on hydration you have.

IT LOOKS LIKE	USUALLY MEANS	DO
A semi-formed stool, sometimes with fur or bone, and a white chalky urate	Normal for a snake eating whole rodents	Nothing.
Urate yellow, orange, or gritty and crystalline	Dehydration at minimum, and in a serious case a urate or kidney problem	Check the water bowl and humidity. Vet if it persists: the same uric acid chemistry is what causes gout.
Watery or unformed for more than one bowel movement in a row	A classic sign of parasites, such as <i>Entamoeba</i> or coccidia	Vet, with a fresh sample in a sealed bag (page 27).
Bright red blood, or dark and tarry	Fresh bleeding near the vent, or digested blood from higher up	Vet the same day.
Straining, with nothing passed	A cloacal problem or a blockage	Vet promptly; straining can end in a prolapse.
An unusually strong or off smell	Something has changed	Compare against your snake's normal and watch for other signs.

How often is normal

A snake generally doesn't pass stool until a meal is mostly digested, so its droppings run on the feeding cycle, not a daily one, and a big meal means a longer gap. The number that matters is your own snake's baseline. A clear gap beyond it, or a sudden change in either direction, is the signal, not any fixed count. Log the date of each one on the Owner Log.

Hydration

- **Urates first.** Smooth, white urate generally means a well-hydrated snake. Dry, hard or gritty urate is the signal to act on.
- **The water bowl is not decoration.** Heavy, big enough to soak in, changed daily and scrubbed weekly. A terrestrial snake that shows dry urates gets easier access to more water.
- **Humidity is part of hydration.** Hold the 55 to 65% baseline. Air that runs too dry tends to show up weeks later as a bad shed.

Reptiles hide illness well. A change in the droppings often shows weeks before the snake looks sick.

Setup checklist & targets

Essential setup checklist

- ☐ 4×2×2 ft (120×60×60 cm) enclosure for an adult, secure and locking, PVC preferred (page 6)
- ☐ Glass only if three sides are covered
- ☐ Heat source: under-tank heater, heat tape, ceramic heat emitter or radiant panel (page 7)
- ☐ A thermostat for every heat source, its probe on the floor of the warm hide on a clip or suction cup (page 8)
- ☐ Digital thermometers and a digital hygrometer, plus an infrared temperature gun
- ☐ Timer for a 12-hour light, 12-hour dark day
- ☐ Cypress mulch or coconut coir, 3 to 4 in (7.5 to 10 cm) deep (page 10)
- ☐ Two snug hides, warm side and cool side, and a humid hide with damp sphagnum moss
- ☐ Heavy, tip-resistant water bowl big enough to soak in
- ☐ Cork bark, a braced branch, leaf litter and plants for cover
- ☐ Frozen-thawed rodents, feeding tongs, and a scale that reads in grams (page 12)
- ☐ A plastic tub for cleaning gear, never the kitchen sink (page 11)

Temperature & climate targets

TARGET	°F	°C
Warm side	88 to 92	31 to 33
Air ceiling, anywhere in the enclosure	never above 95	never above 35
Cool side	75 to 80	24 to 27
Nighttime low	72 to 75	22 to 24
Humidity	55 to 65% baseline, 70 to 80% through a shed	
Light cycle	12 hours light, 12 hours dark, on a timer; no light at night	
UVI, if you add UVB	0.7 to 1.0 across the gradient, 1.1 to 3.0 at the closest point	

Before your ball python comes home

- ☐ Enclosure set up and running, with stable readings, before the snake arrives
- ☐ Warm side, cool side and humidity confirmed on instruments, not by feel
- ☐ Thermostat tested: watch it cut the heat as the probe warms
- ☐ Lid lock tested, and frozen prey in the freezer at the size the seller used
- ☐ If another reptile lives here, a separate quarantine room ready
- ☐ Reptile vet found and the first exam booked for within 72 hours (page 20)

Emergency & quick targets

CALL A REPTILE VET NOW IF

Wheezing, clicking, open-mouth breathing, or mucus or bubbles at the mouth or nose · belly scales spreading discoloration or blistering · pus, pinpoint bleeding or a sour smell in the mouth · anything protruding from the vent · visible mites · a burn · the head held up and back, tremors, or unable to right itself · a meal brought back up · a refusal with weight loss, lethargy or mucus · blood in the stool, or straining with nothing passed

READING	°F	°C
Warm side	88 to 92	31 to 33
Air ceiling, anywhere	never above 95	never above 35
Cool side	75 to 80	24 to 27
Nighttime low, and the power outage floor	72 to 75	22 to 24
Humidity	55 to 65% baseline, 70 to 80% through a shed	
Feeding interval	Every 5 to 7 days to 6 months · 7 to 10 days to 18 months · 10 to 14 days as an adult	
Handling	Not for 48 to 72 hours after a meal, and never during a shed	

Your numbers

Reptile vet _____

Thermostat set point _____

24-hour exotic emergency clinic _____

Last full substrate change _____

Snake's name and hatch date _____

Last vet visit _____

HAVE THIS READY WHEN YOU CALL

Warm and cool side temperatures, current humidity, whether every heat source runs on a thermostat, the date and size of the last meal and whether it stayed down, the date and quality of the last shed, and the weight trend from the Owner Log. Transport in a ventilated tub with a towel, kept warm (page 37).

Budget & shopping list

United States retail ranges, last checked September 2026. The low end is basic but correct gear; the high end is premium PVC and decor.

INITIAL SETUP	TYPICAL COST
4×2×2 ft enclosure for an adult (PVC at the high end)	\$100 to \$450
Heat source: under-tank heater, heat tape, ceramic heat emitter or radiant panel	\$30 to \$80
Thermostat	\$25 to \$100
Thermometers and hygrometer	\$25 to \$50
Substrate, cypress mulch or coconut coir	\$10 to \$40
Hides, water bowl, and cover and decor	\$30 to \$120
Equipment, the budget pick to premium in every row	\$220 to \$840
First vet exam	\$50 to \$150
All-in before the snake arrives	\$270 to \$990
Optional: low-level UVB and a photoperiod LED	\$20 to \$70

Most full setups land at \$300 to \$800, up to \$1,200 with premium PVC and a full bioactive-style build. The snake itself is extra: \$40 to \$100 for a standard or wild-type, \$50 to \$150 for a common single-gene morph, \$250 to \$5,000 or more for designer morphs, and \$25 to \$150 for most rescues. The infrared gun, tongs and scale are not priced separately.

YEARLY ONGOING	TYPICAL COST
Frozen-thawed rodents	\$100 to \$200
Substrate replacement	\$40 to \$120
Electricity, heat held by the thermostat around the clock	\$20 to \$80
Yearly wellness exam	\$50 to \$150
Yearly ongoing, about \$18 to \$46 a month	\$210 to \$550

PLAN FOR, DON'T COUNT ON AVOIDING

A sick or emergency visit runs \$100 to \$500 or more, with imaging or surgery higher. A few hundred dollars set aside matters as much as any item on this list.

WHERE NOT TO CUT CORNERS

The thermostat and the enclosure. The thermostat is the difference between a heat source and a burn, and the enclosure's size and material drive the humidity that most of the health section depends on. Save on decor before you save on either.

First 30 days

Day 1

- ☐ Weigh the snake in grams and photograph it and the setup for a baseline
- ☐ Confirm warm side, cool side and humidity on real instruments, then leave the snake alone
- ☐ Book the first exam with a reptile vet for within 72 hours, with a fresh stool sample (page 20)
- ☐ If another reptile lives here, start the quarantine log (page 40)
- ☐ Fill in the emergency card (page 32)

Weeks 1 to 2

- ☐ Hands off: fresh water and spot cleaning only, no handling and no rearranging (page 17)
- ☐ Offer food on the schedule for its age, at the prey type and size the seller used. A refusal while it settles is ordinary (page 15)
- ☐ Check the thermostat probe hasn't moved at every spot clean

Weeks 2 to 3

- ☐ Begin short handling sessions only once it is eating on a normal schedule
- ☐ Never within 48 to 72 hours of a meal, and not at all once the eyes go blue

Week 4

- ☐ Start the Owner Log with the next weigh-in (page 39)
- ☐ Check the humid hide is still genuinely damp and top up the moss
- ☐ Do the first monthly check of the thermostat and probe
- ☐ Introduce the first enrichment item (page 41)

Days 31 to 90

- ☐ Book any follow-up the vet recommended, and act on the fecal result
- ☐ Watch the first shed closely and log it, including whether both eye caps came off (page 29)
- ☐ Look back at the Owner Log for any slow drift in weight, appetite or humidity
- ☐ Do the first full substrate change: monthly if messy, at least every 3 months (page 11)
- ☐ Fill in the pet-sitter sheet before you need it (page 38)
- ☐ Keep any quarantine running the full 3 to 6 months before the snake shares a room, tools or air with another reptile

Symptom quick reference

SYMPTOM	LIKELY CAUSE	ACTION
Wheezing, clicking, open-mouth breathing, bubbles at the mouth	Respiratory infection	Vet promptly; warm side to target (page 25)
Belly scales discolored, soft or blistered	Scale rot	Clean, dry enclosure now; vet if spreading (page 25)
Pus, sour smell or pinpoint bleeding in the mouth	Mouth rot	Vet; it does not resolve alone (page 26)
Specks on the scales or in the water bowl, long soaking	Mites	Vet, plus a full enclosure deep clean (page 26)
A burn	Unregulated or reachable heat	Fix the heat source; vet (page 26)
Weight loss while eating normally	Internal parasites	Vet with a fresh stool sample (page 27)
A meal brought back up	Regurgitation	Skip a feeding cycle and call the vet (page 27)
Head up and back, tremors, can't right itself	Possible inclusion body disease	Vet now; isolate from other snakes (page 28)
Anything protruding from the vent	Prolapse	Keep moist, don't push back, vet now (page 28)
Shed in patches, or eye caps left on	Low humidity, or one of five other causes	Soak for stuck skin; vet for eye caps (page 29)
Yellow, orange or gritty urates	Dehydration	Water bowl and humidity; vet if it persists (page 30)
Watery stool more than once in a row	Parasites	Vet with a fresh sample (page 30)
Refusing food, otherwise normal and holding weight	Setup, shed, stress, or season	Work the list in order (page 15)
Refusing food from fall into winter	The seasonal appetite dip	Weigh every 1 to 2 weeks (page 22)
Refusing food with weight loss, lethargy or wheezing	Several possible causes	Vet, and bring the Owner Log (page 24)
Triangular cross-section, ridge down the spine	Underweight	Vet, rather than just more food (page 16)
Round body, skin folds along the sides	Overweight	Adjust prey size or interval, never skip meals (page 16)
Balled up tight, head tucked in	Fear, normal defensive pose	Leave it be (page 18)

Daily, weekly & seasonal routine

Daily

- ☐ Read the warm-end and cool-end thermometers and the hygrometer; a warm end that keeps climbing is the warning sign
- ☐ Change the water
- ☐ Spot clean as soon as waste appears, and look at it first (page 30)
- ☐ Quick visual check: breathing, mouth, eyes, belly scales
- ☐ Wash your hands after every contact

Every feeding

- ☐ Feed on the interval for its age, and log the prey and whether it was eaten (page 12)
- ☐ No handling for the next 48 to 72 hours

Weekly

- ☐ Scrub the water bowl and any soiled decor; 3% bleach left on 10 minutes, rinsed and dried (page 11)
- ☐ Check the humid hide is damp, and the probe is still on the floor of the warm hide

Monthly

- ☐ Weigh a growing snake and log it (page 39)
- ☐ Confirm the thermostat still cuts the heat at its set point
- ☐ Full substrate change if the snake is messy
- ☐ Review the Owner Log for slow trends, not just single readings

Every 3 months

- ☐ Full substrate change at the latest, enclosure disinfected and dried
- ☐ Check the hides still fit snugly as the snake grows

Seasonal

- ☐ Re-check the room in winter: the cold-weather drop is when respiratory infection spikes
- ☐ Expect less appetite from roughly October to March; weigh every 1 to 2 weeks while it lasts (page 22)

Yearly

- ☐ Wellness exam with a reptile vet (page 24)
- ☐ Measure the snake against the enclosure: it should be at least as long as the snake
- ☐ Replace an optional UVB tube on the maker's schedule, 6 to 12 months

Power outages, heat waves, travel & transport

A healthy adult tolerates one cool night without harm. The risk is days without heat, and feeding a snake that can't digest.

When the power goes out

Do

- Keep the snake at or above its night range, 72 to 75°F (22 to 24°C). Below it for more than a day or two is the classic winter trigger for respiratory infection. A battery thermometer in the enclosure tells you the real number.
- Drape thick towels or a blanket over the enclosure, leaving a vent gap.
- Move the enclosure, or the snake in a ventilated tub, to the warmest room, usually one without outside walls.
- Use a wrapped hot-water bottle or hand warmers beside the snake, never touching it.
- Stop feeding. Water only until heat is reliably back.
- Past a day or two below the floor, board it with a vet, a friend with power, or a reptile shop.

Don't

- Don't use candles, gas stoves or open flame anywhere near the enclosure: a fire and carbon monoxide risk.
- Don't put a heat pack in direct contact; a cold, sluggish snake can't move away fast enough and will burn.
- Don't feed "to keep its strength up." Undigested food in a cold gut is a bacterial risk, not a benefit.
- Don't warm a cold snake in hot water: lukewarm only, and only once it is alert.

OUTAGE KIT

Hand warmers or a hot-water bottle, two thick towels, a battery thermometer, a ventilated locking tub, and the vet numbers on paper.

In a heat wave

The air anywhere in the enclosure must stay at or below 95°F (35°C). A thermostat can cut the heat source but cannot cool the room, so read both ends through the day while the heat lasts.

Leaving town

TRIP LENGTH	ADULT	JUVENILE, SICK OR MID-SHED SNAKE
1 to 2 nights	Often fine alone. Thermostat running, light on its timer, fresh water before you go.	A sitter once a day for water and a temperature check.
3 to 7 nights	A sitter every 1 to 2 days for water, a stool check and a temperature check. Feed on the normal schedule if a meal falls in the trip.	A sitter daily, with the pet-sitter sheet filled in.
Over a week	A daily sitter who has done a practice visit with you, or boarding with a reptile-experienced vet or shop. Never a general dog-and-cat boarding facility.	

Transport and vet trips

- A ventilated, locking plastic tub with a towel to grip, inside a dark, room-temperature bag or box. No loose substrate, no water dish.
- Warm the car first in cold weather, and add a wrapped hand warmer beside the tub for trips over 30 minutes below 65°F (18°C). Check the tub every 20 to 30 minutes at a stop.
- Don't feed right before a trip. Never leave the snake in a parked car, hot or cold, even briefly.
- Moving house: set the enclosure up and running at the new place before the snake goes in, the same as day one.

FILL IN AND LEAVE BY THE ENCLOSURE

Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it once in person before the first solo visit.

Snake's name _____

Age and normal weight _____

My phone _____

Reptile vet _____

24-hour emergency clinic _____

Backup contact _____

Each visit

- ☐ Check the thermometers. Warm end should read 88 to 92°F (31 to 33°C), cool end 75 to 80°F (24 to 27°C), humidity 55 to 65%. The light is on between _____ and _____.
- ☐ Rinse and refill the water bowl. It's normal for the snake to be sitting in it now and then.
- ☐ Find the snake and check it's breathing quietly with its mouth closed. It will usually be in a hide or under cover, which is normal.
- ☐ Scoop any waste with a paper towel. Normal is a dark stool with a white, chalky end, not every day.
- ☐ Feed only if I've marked a date: _____. Prey is in the freezer, size _____. Thaw and warm it per the note on the fridge, offer it with the tongs, and take it out if it isn't eaten.
- ☐ Wash your hands before and after. Don't rinse anything in the kitchen sink.

Please don't

- Take the snake out or handle it, unless we agreed on it. It is fine on its own for the whole trip.
- Change any dial, timer or thermostat setting, or move anything in the enclosure.
- Feed anything not on this sheet, and never a live animal. Never microwave the prey.
- Leave the lid unlatched, even for a moment. This snake will find any gap.

CALL ME IF

Any reading below 72°F (22°C) or above 95°F (35°C) · the light is off during the day · the snake is breathing with its mouth open, wheezing or clicking · there are bubbles or mucus at its mouth or nose · there are tiny specks on the snake or floating in the water bowl · its belly scales look discolored or blistered · anything is protruding from the tail end · it has brought a meal back up · it hasn't moved from the same spot across two visits · you can't find it at all · anything else that looks wrong. If I don't answer, call the vet number above.

One row per feeding or weigh-in. Photocopy this page, or track digitally. Consistency matters more than the format.

WHY BOTHER LOGGING

In a species that normally refuses meals, the weight trend is what tells you whether a fast is routine or a symptom. Most of the conditions in this package show up as a weight change before anything else is visible, and a trend line tells a vet more than a single reading.

Equipment, quarantine & vet log

Equipment log

ITEM	INSTALLED	REPLACE OR CHECK BY	DONE
Thermostat and probe check (monthly)			
Heat source			
Full substrate change (monthly if messy, every 3 months at most)			
Thermometer and hygrometer batteries			
Optional UVB tube (6 to 12 months)			

Twelve-month planner

Tick the box in the month you did it. Month 1 is whichever month you start in.

TASK, BY MONTH	1	2	3	4	5	6	7	8	9	10	11	12
Thermostat check	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Full substrate change	☐			☐			☐			☐		
Vet exam	☐											

Quarantine log (any new snake, 3 to 6 months)

DATE IN	DATE OUT	WEEKLY WEIGHT, MITE CHECKS, FEEDS, SHEDS, ANYTHING NOTICED

Vet visit log

DATE	WEIGHT (G)	REASON AND FINDINGS	FOLLOW-UP

Enrichment checklist & log

Setup

- ☐ Cover across the whole floor, not only inside the two hides
- ☐ Stable climbing structure braced at both ends, low enough that a fall isn't one
- ☐ A water basin big enough for the snake to fully enter
- ☐ Living plants and leaf litter breaking up sightlines

Monthly review

- ☐ Add or shift one piece of cork or a branch, without blocking the warm spot
- ☐ Offer a scent: a shed skin from another enclosure, or prey dragged across the substrate before feeding
- ☐ Check the snake uses more of the floor than last month, not less

Enrichment log

DATE	ACTIVITY	HOW IT WENT, AND WHETHER TO REPEAT

WHAT NOT TO DO

Don't treat handling as enrichment, especially around a shed. Don't stack decor so densely that the snake can't reach the warm spot: thermal access outranks everything here. And don't call a bare enclosure enriched because one item went into it: the housing study on page 19 compared whole setups, and the bare one lost.

REFERENCE

Glossary, A to L

ASF	African soft-furred rat: a different rodent species from the common feeder rat, milder-smelling and thinner-skinned. Used to restart a stubborn feeder, not as a diet. Page 13.
Brumation	A reptile's seasonal slowdown in activity, appetite and metabolism, tied to temperature and daylight. Not true hibernation: the snake can still wake, move and drink. Page 22.
Captive-bred	Hatched from parents kept by people rather than taken from the wild. The better pet: less likely to bite, refuse food or carry parasites. Page 5.
CBS	Columbia Broadcasting System: a United States television network whose news service reported the 62-year-old zoo female in this package's sources.
Ceramic heat emitter	A screw-in heater that gives off heat with no visible light, so it can run at night. Page 7.
CITES	Convention on International Trade in Endangered Species of Wild Fauna and Flora: the treaty that tracks international trade in species that trade could threaten. Ball pythons are listed on its Appendix II. Page 5.
Cloaca	The single opening under the tail that waste, urates and eggs all leave through. The vent is its outer opening. Page 17.
Cloacal spurs	Small claw-like points either side of the vent in boas and pythons. A real sex-linked trait in some populations, but no way to sex a single snake. Page 21.
<i>Cryptosporidium serpentis</i>	A gut parasite that thickens the stomach lining and causes repeated regurgitation. No reliably effective treatment, and can be fatal. Page 27.
Deep heat projector	A heater like a ceramic emitter that sends its heat down in a more focused beam, with no visible light. Page 7.
Dimming thermostat	A thermostat that turns the power to a heat bulb down instead of switching it off and on. The right kind for halogen and incandescent bulbs. Page 8.
Dysecdysis	Incomplete or "stuck" shedding, where the skin comes off in pieces. Low humidity is the most common single cause, one of six. Page 29.
Egg binding	A female unable to pass her eggs. One of the causes of straining hard enough to push tissue out of the vent. Page 21.
Eye cap	The clear scale over each eye, also called the spectacle, shed with the rest of the skin. A retained cap is easy to miss and is a vet's job to remove. Page 29.
Fecal test	A vet's microscope check of a stool sample for parasites. Bring a fresh sample in a sealed bag. Page 20.
Ferguson Zone	One of four groups that match a species' natural sun exposure to a UVB strength. Ball pythons are Zone 2, partial-sun and occasional baskers. Page 9.
Frozen-thawed	Prey frozen, then thawed and warmed before feeding. The standard diet, and far safer than live prey. Page 14.
Gravid	Carrying eggs. Used for reptiles in place of pregnant. Page 5.
Heat pits	Openings along the upper lip that detect the body warmth of prey, letting a ball python strike accurately in total darkness. Why prey is warmed before feeding. Page 14.
Hemipenes	A male snake's paired reproductive organs, held inverted in pockets at the base of the tail. Probing and popping both work by finding them. Page 21.
Humid hide	A closed hide packed with damp sphagnum moss: a humid pocket the snake can enter and leave, kept year round. Page 9.
Hygrometer	A humidity gauge. Use a digital one; analog dials are notoriously inaccurate. Page 9.
IBD	Inclusion body disease: a fatal virus of boas and pythons, caused by reptarenaviruses, that shows as nerve signs such as stargazing. No cure; quarantine is the only defense. Page 28.
Infrared temperature gun	A handheld tool that reads the temperature of a surface from a distance. How you check what the snake actually lies on. Page 7.
LED	Light-emitting diode: a low-heat, long-life lamp. Used on a timer for the 12-hour day. Page 9.

REFERENCE

Glossary, M to Z

Morph	A selectively bred line with a different color or pattern. Some genes carry a health cost, such as the spider morph's wobble. Page 5.
Mouth rot	Infectious stomatitis: an infection of the gums and mouth, usually riding on stress or poor husbandry. Needs a vet. Page 26.
Musking	A foul-smelling release from the cloaca. A clear sign the snake feels seriously threatened and the session is over. Page 17.
Nematodes	Roundworms. Some turn up often in snake stool and range from minor to a cause of gut blockage or pneumonia. Page 27.
On/off thermostat	A thermostat that cuts power fully at the set point and restores it below. Suits an under-tank mat. Page 8.
PCR	Polymerase chain reaction: a lab test that detects a germ's genetic material. Used with a biopsy to diagnose IBD. Page 28.
Probe	The thermostat's sensor. On the floor of the warm hide, held with a clip or suction cup, never tape. Page 8.
Probing and popping	The two hands-on ways to sex a snake. Both are for a vet or an experienced breeder; done wrong, either can injure the snake. Page 21.
Prolapse	Internal tissue pushed out through the vent. Keep it moist, never push it back, and see a vet at once. Page 28.
Protozoa	Single-celled creatures. Most found in reptile stool are harmless; a few, such as <i>Entamoeba invadens</i> , cause serious illness. Page 27.
Pulse proportional thermostat	A thermostat that sends short pulses of power to hold a steady temperature. Suits a ceramic heat emitter. Page 8.
PVC	Polyvinyl chloride: a rigid, waterproof plastic used for enclosure panels. Holds humidity far better than glass. Pages 6, 33.
Quarantine	Keeping a new snake fully apart from other reptiles, in a separate room with separate tools, for 3 to 6 months. Page 20.
Ranching	Collecting eggs and gravid females from the wild, raising the hatchlings briefly and exporting them. How most wild-sourced ball pythons now reach the trade. Page 5.
Regurgitation	A meal eaten and then brought back up. Stop feeding for a cycle and call the vet. Page 27.
<i>Salmonella</i>	Bacteria that healthy reptiles carry in the gut and shed in droppings without being ill. Spread from hands to mouth. Page 11.
Scale rot	Infectious dermatitis of the belly scales, driven by damp, dirty substrate. Page 25.
Stargazing	The head held up and back at an odd angle, as if staring at the sky. A nerve sign, the mark of IBD, and an emergency. Page 28.
Units	°F, degrees Fahrenheit, and °C, degrees Celsius, for temperature; in (inches), ft (feet) and cm (centimeters) for size; g (grams) for weight; % (percent) for humidity.
Urate	The white, chalky part of reptile waste, made from uric acid instead of liquid urine. Its color and texture are a hydration gauge. Page 30.
UVB	Ultraviolet B: the part of sunlight many reptiles use to make vitamin D3. Optional for ball pythons, which have been kept for decades without it. Page 9.
UVI	Ultraviolet index: a measure of UVB strength at one spot. 0.7 to 1.0 across the gradient and 1.1 to 3.0 at the closest point, if you add UVB. Page 9.
VCA	Veterinary Centers of America, now VCA Animal Hospitals: an animal hospital chain whose pet care articles are among this package's sources.
Wobble syndrome	A heritable nerve disorder linked to the spider morph: head wobbling, poor coordination, and trouble striking or constricting prey. Page 5.

REFERENCE

Sources

Where the numbers in this package come from, so you can check them yourself.

Veterinary references

- Merck Veterinary Manual: management and husbandry of reptiles; nutrition in snakes; parasitic, viral and bacterial diseases of reptiles; disorders and diseases of reptiles; routine health care; selecting a reptile; clinical procedures; important husbandry requirements; overview of reptiles. The 3 to 6 month quarantine, thermostat control, IBD in boas against pythons, and the parasite findings.
- VCA Animal Hospitals: feeding pet snakes; snakes, housing; owning a pet snake; veterinary care, new reptile visit; choosing a pet reptile; reptiles, gout. The 4 to 6 in heater clearance, the first exam, and the pre-purchase checks.
- PetMD, Ball Python Care Sheet, by veterinarian Maria Zayas. Association of Reptile and Amphibian Veterinarians: the find-a-vet directory.
- Kevin Wright, veterinarian, in Reptiles Magazine: retained shed, the 30-minute soak and the six causes. Margaret Wissman, veterinarian, in Reptiles Magazine: snake winter anorexia. WikiVet: snake shedding. Purdue University College of Veterinary Medicine: shedding in reptiles. Vet Times: cloacal organ prolapse in reptiles.

Research

- Kaufmann and colleagues, PLOS ONE: animal-appropriate housing of ball pythons, a behavior-based evaluation of two housing systems. Hutchings (2025), International Journal of Zoology: the spatial needs of captive snakes. Burghardt (2013): environmental enrichment and cognitive complexity in reptiles and amphibians. Dropping the Ball? The welfare of ball pythons in the pet trade, Animals.
- Rose and Williams (2014), Journal of Exotic Pet Medicine 23(3): neurological dysfunction in a ball python color morph. Starck, Schrenk, Schröder and Pees (2022), PLOS One: malformations of the inner ear in spider morph pythons.
- Brashears and DeNardo (2015), Journal of Comparative Physiology A: heat production during brooding is not the norm among pythons. Vetere, Capasso and Di Ianni (2025), Animals: sex determination in reptiles. Hoefer, Robinson and Pinou (2021), Herpetology Notes: pelvic spurs of the Bahamian boa. Falk, Snow and Reed (2017), PLOS ONE: body-condition indices in a giant snake. Holthaus and colleagues (2017), Scientific Reports: the skin of snakes.

Public health

- Centers for Disease Control and Prevention: Healthy Pets, Healthy People, reptiles and amphibians; advice to pet owners on *Salmonella* from frozen feeder rodents. Center for Food Security and Public Health, Iowa State University: reptile- and amphibian-associated salmonellosis, and isolation and quarantine areas. United States Food and Drug Administration: storing food safely. Merck Veterinary Manual: cleaning and disinfection.

Husbandry, prices and the law

- ReptiFiles: ball python enclosure size and lighting, substrate, respiratory infection, and finding a lost snake; its care sheet as republished by Zen Habitats. Reptile Direct: ball python care. North Hampton Animal Hospital and the Wisconsin Herpetological Association: escaped snakes. Spyder Robotics, thermostat manual: probe placement and the warning against tape. National Geographic, and CBS News (the Columbia Broadcasting System's news service): the export trade and the 62-year-old zoo female.
- Prices checked in September 2026 at Zen Habitats, Josh's Frogs, PetSmart and Colorado Exotic Animal Hospital, with electricity from the United States Energy Information Administration.
- Federal Register 2012-1155, the injurious wildlife listing; United States Fish and Wildlife Service on the 2017 court ruling; Florida Fish and Wildlife Conservation Commission, Reptiles of Concern. Hawaii Administrative Rules § 4-71-6; New York City Health Code § 161.01; New Jersey Administrative Code 7:25-4.3; 3 Delaware Administrative Code 903; Minnesota Statutes § 97A.401; West Virginia Code § 20-2-13; and each other state's rules, read on 1 October 2026 (Maine on 5 August 2026).
- Beastly Facts guides, current at publication: the ball python enclosure setup, feeding, handling, health issues, enrichment, cost and legal guides and the two ball python comparisons, the ball python encyclopedia profile and fun facts, and the cross-species reptile heating and thermostats, quarantine, *Salmonella* and hygiene, stool and urates, shedding, snake shedding, snake brumation, snake sexing and body condition, ultraviolet lighting and emergency plan guides.

REFERENCE

Where the sources disagree

Where credible sources give different numbers, this package prints one and says why.

TOPIC	WHAT THE SOURCES SAY	WHAT THIS PACKAGE USES
Adult length	3 to 5 ft (care references and the species profile); 5 to 6 ft at maturity, reached at 3 to 5 years (a veterinary hospital chain)	3 to 5 ft, females significantly larger
Longest-lived	Up to 48 years recorded in captivity (species profile); a zoo female laying eggs at 62 (news reports)	20 to 30 years typical, some far longer
Conservation grade	Least Concern (species profile); Near Threatened (a state legal record)	No grade printed; the export pressure is stated instead
Weight by age	Care sheets print ranges; no veterinary or peer-reviewed source gives verified snake weights by age	None printed: read the trend
Quarantine	3 to 6 months (veterinary); 60 to 90 days (a hobby floor); two to four weeks (informal advice)	3 to 6 months
Full substrate change	Monthly if messy, at least every 3 months (species setup guidance); bark or mulch every few weeks (a general reptile cleaning schedule)	The species figure
Night heat	A snake needs none in a room at 65 to 70°F (general snake guidance); a 72 to 75°F night range (species guidance)	72 to 75°F; night heat if the room falls below it
Thermostat probe under an overhead heater	Makers split between the surface under the heater and the air a few inches below it	The controller's own manual, at the snake's level
Stuck shed soak	Chin-deep water at enclosure temperature for 30 minutes (a reptile veterinarian); warm water for several hours (a veterinary manual)	30 minutes, supervised; a vet for anything that won't come away
Prey warmth	Room temperature or warmer (veterinary); "lukewarm" (a care sheet); no source gives a number	Warm to the touch, judged by hand
Yearly running cost	\$200 to \$500 a year, or \$20 to \$60 a month (summary figures); \$210 to \$550 (the itemized rows, with the yearly exam)	The itemized rows, so the total adds up
Where legal	A permit in New Jersey and Delaware (a care summary); New Jersey, Delaware and West Virginia (the state rules)	The state rules: three permit states

REFERENCE

Version history & about

What changed in each edition, so a printed copy can be placed.

Version history

This is edition 3.0 (Oct 2026). The store page shows the current edition; re-download free if yours is older. Where this list and a page disagree, the page wins.

- **3.0, Oct 2026.** Rebuilt on the current outline, 46 pages. Safety: a refusal is checked in order, temperature and humidity first, then a shed, stress, breeding season (October to March) and the cooler-month slowdown; thermostat probe on the floor of the warm hide, held with a probe clip or suction cup, never tape; first vet exam within 72 hours, no later than two weeks; the shed follows about 4 to 7 days after the eyes clear; stuck shed loosened with a 30-minute chin-deep soak at enclosure temperature; after a regurgitation, skip a feeding cycle and call the vet; prey sized by width alone, with no body-weight percentage; an overweight snake gets its prey size or interval adjusted, never meals skipped; the water bowl and soiled decor scrubbed weekly with 3% bleach left on 10 minutes, and the whole substrate changed monthly if messy and at least every 3 months; brooding females do not warm their eggs by shivering; weights by age removed, since no verified source gives them. Legal: a new page on the law in all 52 jurisdictions, barred in Hawaii and New York City, a permit in New Jersey, Delaware and West Virginia, and a condition in Minnesota. Costs: equipment \$220 to \$840 and running costs \$210 to \$550 a year, from the itemized prices. Other changes are wording only.
- **2.2, Sep 2026.** Safety: quarantine for a new snake 3 to 6 months; first vet exam within two weeks, with a fecal sample; a second thermometer read daily; any overhead heater at least 4 to 6 in above the snake's reach; prey warmth judged by touch, with no core temperature; frozen prey never refrozen; *Salmonella* precautions widened to adults over 65, with children under 5 kept from reptiles and their enclosures. Other changes are wording only.
- **2.1, Sep 2026.** Safety: baseline humidity 55 to 65%, down from 55 to 70%; 70 to 80% through a shed. Other changes are wording only.
- **2.0, Sep 2026.** Rebuilt, 34 pages, with substrate and handling on their own pages and the full handling guidance restored.
- **1.0, Sep 2026.** First edition, 33 pages. Replaced by 2.0.

This package is a husbandry reference compiled from the listed sources and cross-checked against current veterinary guidance at the time of publication. It is not a substitute for a reptile-experienced veterinarian, and nothing in it should delay a vet visit. Treatments described here are for recognizing what a vet may do, never for use at home. The law page is a summary as of October 2026, not legal advice.

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